

Ghulam Mustafa

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EDUCATION

Habib University Bachelor of Science in Computer Science Awarded 20% merit-based scholarship	Karachi, Pakistan 2022 – 2026 (Expected)
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RESEARCH EXPERIENCE

Web-based GIS Applications (Habib University) Project: Weather Anomalies Dashboard - Developed GIS applications using ERA5 climate datasets spanning multiple decades to visualize district-level temperature and rainfall anomalies across Pakistan. - Implemented an interactive GIS application that performs real-time anomaly detection by comparing current weather data against historical district-level averages. - Currently developing statistical baseline forecasting approaches based on historical climate patterns to estimate future anomaly trends.	May 2025 – Present
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PROJECTS

Safe Karachi: Intelligent Urban Safety Platform (FYP) Tech Stack: React Native, Node.js, PostgreSQL/PostGIS - Architecting a cross-platform mobile application that calculates secure commuter routes by integrating crowdsourced incident reports and social media data. - Developed a crime hotspot prediction model using XGBoost/Random Forest to dynamically score road safety across Karachi. - Developing a routing algorithm that balances physical distance with real-time safety metrics to provide secure navigation for city commuters.	Aug 2025 – Present
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PAN 2026 Lab: AI Detection & Generative Plagiarism Detection - Designed a hybrid AI detection pipeline (Task 1) combining fine-tuned DeBERTa-v3-base CLS embeddings with statistical surprisal features from a frozen GPT-2 model, fused into a 1033-D vector and classified via XGBoost; achieved ROC-AUC of 0.9999 and F1 of 0.9968, up from 0.981 and 0.943 respectively. - Built an adversarial data augmentation pipeline using five LLMs (Llama 3, Mistral, Qwen, Gemma, Phi-4) to humanize AI-generated text, improving robustness against obfuscated and style-mimicking AI authors. - Developed a two-stage hybrid retrieval system for generative plagiarism detection (Task 4) using BGE-M3 embeddings with dense + sparse similarity, followed by a BGE Reranker v2-m3 for chunk-level re-scoring; achieved nDCG@10 of 0.3919 and Recall@10 of 0.3974, more than doubling the BM25 baseline (0.1865 and 0.1886).	Jan 2026 – Present
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AutoAnnotate: Automatic Annotation for Urban Navigation - Built a zero-shot automatic annotation pipeline for South Asian urban road scenes by chaining GroundingDINO (open-vocabulary detection), SAM2 (pixel-level segmentation), and CLIP (label verification) to produce COCO-format annotations without any manual labeling. - Created a custom class list with synonym-based prompts to improve GroundingDINO's recall on South Asian vehicle types absent from western datasets. - CLIP verification stage boosted annotation F1 from 0.4738 to 0.7376, driven by a recall improvement from 0.3497 to 0.6863, and auto-generated labels were used to train a downstream YOLOv8 detector evaluated against a human-annotation baseline.	Jan 2026 – May 2026
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TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, JavaScript, C#, SQL
Machine Learning & AI: PyTorch, TensorFlow, HuggingFace, OpenCV, Pandas
Backend & Frontend: Node.js, Express.js, Spring Boot, React Native, React.js
Tools & Platforms: Git, GitHub, Google Earth Engine, PostgreSQL/PostGIS